



FRANKONIA
GROUP

CyberShield

**Next-Generation Electromagnetic Shielding
& High-Security Data Center Enclosures
& Home Protection Solutions**

A measureable electromagnetic boundary.
Engineered, integrated and verified as one system.





The Frankonia Group was founded in 1987 as a solution provider for EMC and antenna test laboratories, meeting the increasing demand for highly specialized testing environments for the electronic testing and protection industry.

Without limitations in its capabilities, Frankonia develops future-oriented concepts for our complete product range, which guarantee the optimal use of resources, as well as the best possible customized solutions.

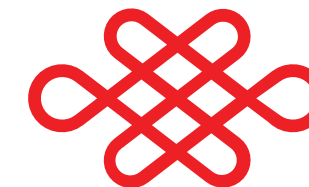
Frankonia offers complete solutions for the electronic, military and automotive industry, which meet customers' individual requirements.

Frankonia **trusted solutions**

- › Turnkey and future-proof solutions
- › Expertise in every stage of a project
- › Extended services: planning & consultation, testing readiness
- › Everything from one source
- › In-house engineering and manufacturing
- › High quality product scope
- › Global presence and worldwide activity

Frankonia CyberShield®

The unique and trustworthy
partner for worldwide
Shielding Solutions.



FRANKONIA

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Frankonia is recognized as a highly specialized technology corporation for EMC shielding solutions within the industrial and military sectors for immunity to electromagnetic disturbances.

Frankonia is the preferred supplier for complete solutions when it comes to the implementation of complex shielded test facilities. Passionate employees, many of them with longtime affiliation and experience, plan, coordinate, and define customized solutions with and for our customers that meet today's and future standards.

With our expertise, flexibility, quality and a high degree of technology, we generate future-proof solutions on a global scale. Due to our commitment to providing holistic solutions, we offer outstanding expertise in every phase of a project.

Frankonia's project business convinces with its in-house project management, engineering and production, a trend-setting research, as well as its own installation and service teams. In this way, we ensure to provide a high level of technology and quality.

Frankonia **at a Glance**

- › Frankonia demonstrates a global presence in cooperation, with a well-structured network of productions, representations and service units.
- › Frankonia provides fundamental knowledge to operate as a complete solution provider.
- › Frankonia implements innovative technologies to enhance efficiency and improve the outcomes and quality along with customers' needs.
- › Frankonia strives to be the preferred partner for customized and state-of-the-art solutions.

Frankonia's Shielding Solutions are part of every high-security facilities worldwide. Customers from the commercial and military industry trust Frankonia's solutions for now more than 35 years.

Frankonia CyberShield®

Integrated Electromagnetic Security & Protection Solutions

One boundary. Designed, built and measured under a single responsibility.

Electromagnetic shielding structure, doors, filters, ventilation waveguides and every penetration are engineered as one continuous boundary, assembled weld-free alongside live operations, then measured on site to EN 50147-1 / IEEE 299 and handed over as documented evidence.

Applications that require performance

In an era defined by ubiquitous surveillance, electromagnetic threats, and critical infrastructure vulnerabilities, protecting mission-critical assets requires absolute defense. Our sovereign-grade shielding technology provides verified, high-performance protection engineered to safeguard data, strategic operations, and human life against physical, electronic, and aerial exposure.

From global compute clusters to secure meeting chambers, our custom shielding solutions deliver measurable, certified security where performance is mandatory.



Industrial Applications

Hyperscale Cloud & AI Centers

- › Core AI compute clusters, quantum hardware and critical availability zones, shielded against RF tampering and high-power electromagnetic threats.

Colocation & Enterprise Data Centers

- › A certified shielded vault offered as a premium, measurable security tier for enterprise customers under strict governance requirements.

Defence, Government & Financial Institutions

- › Sovereign cloud deployments, command centres and high-frequency trading platforms where confidentiality cannot be left to assumption.

Drone Defense & Signature Concealments

- › Anti-drone shielding, aerial surveillance evasion, and shelters where visual and electronic privacy cannot be left to assumption.

Focus: Infrastructure, AI Cloud & Data Centers

- › Signal Isolation for AI & Quantum Infrastructure: Eliminates ambient electromagnetic noise that distorts sensitive quantum state measurements and high-density GPU matrix calculations.
- › Sovereign Data & Perimeter Containment: Guarantees absolute physical isolation of sovereign cloud servers so zero data emissions cross national or institutional boundaries.
- › High-Power Directed Energy & EMP Protection: Shields critical availability zones against high-altitude electromagnetic interference (HEMP) and targeted weapons designed to destroy high-frequency processors.
- › Modular Vault Scale: Uses scalable, retrofittable shielded cages and server rooms that adapt as enterprise tenants expand their physical footprint.

Home Shielding & Family Protection

- › Sovereign-grade disaster prep, private family bunkers, and residential safe rooms where personal security and total confidentiality cannot be left to assumption.

Eavesdropping Prevention & Secure Chambers

- › Acoustic isolation, counter-surveillance shielding, and secure meeting chambers where sensitive conversations and private intelligence cannot be left to assumption.

Focus: Preventive Protection & Private Applications

- › Electromagnetic Resiliency: Protect off-grid energy storage, solar inverters, communication links, and medical equipment against high-altitude electromagnetic pulses (HEMP), coronal mass ejections (CMEs), and directed energy bursts.
- › Perimeter Signal Containment: Eliminate RF/Wi-Fi signal leakage from the residence to prevent external digital reconnaissance, drone surveillance, and wireless perimeter breaches.
- › Physical & Environmental Autonomy: Maintain life-support systems, clean air, and critical communications during grid failures or civil disruptions without exposing external vulnerability signals.

Home Security Applications



Reliable

Next-Gen Shielding Standard

Proven performance

Every CyberShield project is engineered around your facility, and ends with measured evidence. Shielding effectiveness is tested on site according to international standards, and documented acceptance with detailed performance results are handed over.

EN 50147-1

Shielding effectiveness measurement

BSI TL-03305 / 03306

Eavesdropping-protected rooms & IT enclosures

MIL-STD-188-125-1 / -2

HEMP and IEMI protection, project-specific validation

IEEE 299

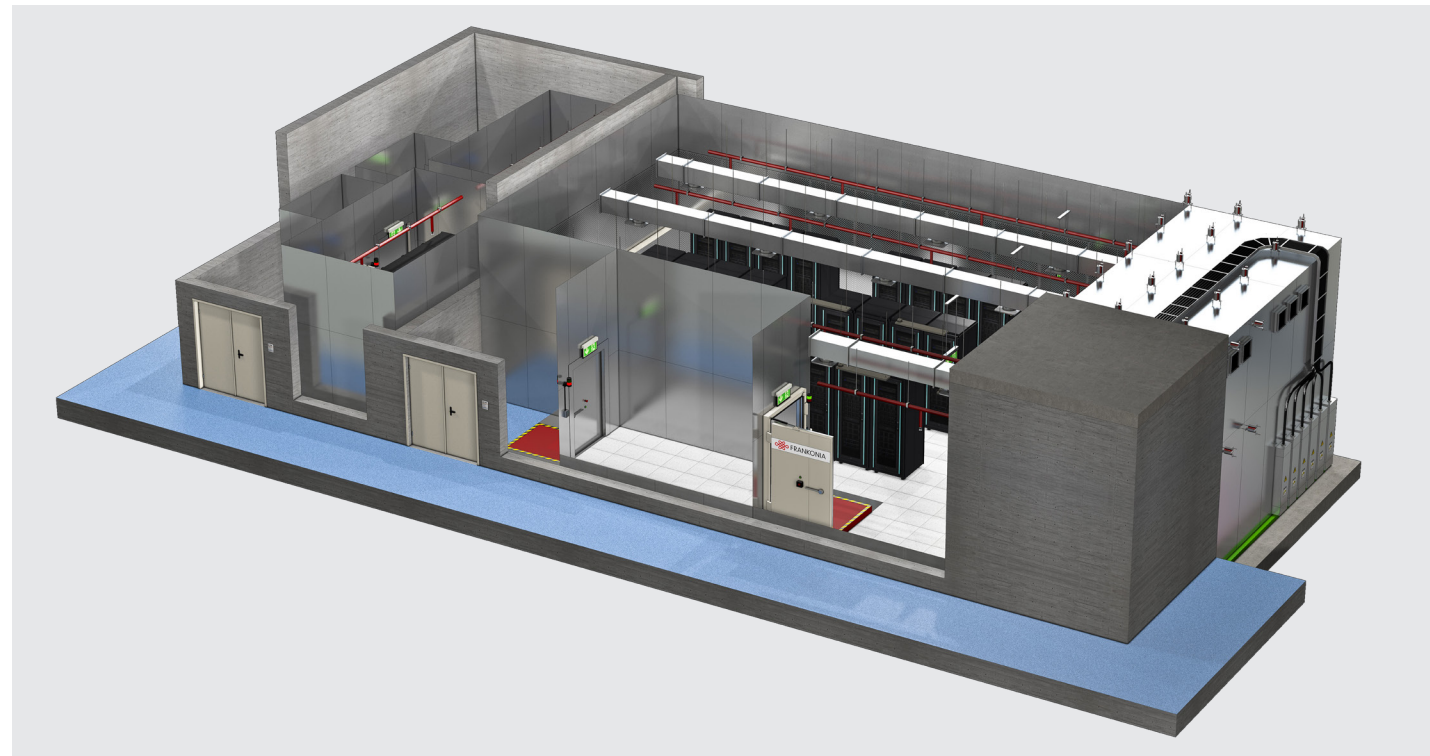
Shielding effectiveness measurement

NATO SDIP-27 Level A

TEMPEST design & approval, aligned with NSA 94-106

ISO/IEC 27001

Supports the physical and environmental security controls



High-Performance Compute Infrastructure

- › Core compute zones and high-density, accelerated server racks.

Cryptographic & Administration Assets

- › Key management systems, cryptographic infrastructure, and privileged access control facilities.

Network Interconnection Points

- › Network entrance facilities, meet-me rooms, and cross-connect nodes.

Critical Power & Auxiliary Infrastructure

- › Power distribution hubs, uninterrupted power supply (UPS) corridors, and all filtered service penetrations crossing the perimeter.

Performance shielding

Prefabricated PAN modules pass through standard building doors, assemble from the inside and can be installed close to existing walls. No glue. No welding. No irreversible commitment.

01 Precision assembly

Panels are bolted every 75 mm with predefined torque and conductive mesh gaskets.

02 Architectural integration

Reversible modules leave flat interior surfaces for finishing walls, ceilings and racks, designed around raised floors, fire systems, lighting, cooling and access control.

03 Adaptable by design

Dismountable without damage for expansion, modification or complete relocation.

04 Complete boundary

Shielding structure, doors, filters, honeycombs and waveguides are treated as one system.

Technology received

- › On-site shielding effectiveness measurement across the specified frequency
- › Leak detection sweep of doors, filters, honeycombs, and every feed-through
- › Documented acceptance report issues with the room
- › Periodic re-testing and security re-certification as service

>90 dB

at 10 kHz
magnetic field

>120 dB

at 100 MHz
plane wave

>100 dB

at 40 GHz
microwave



Trusted

Shielded Environment

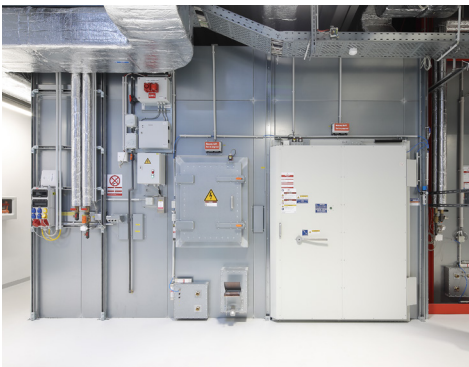
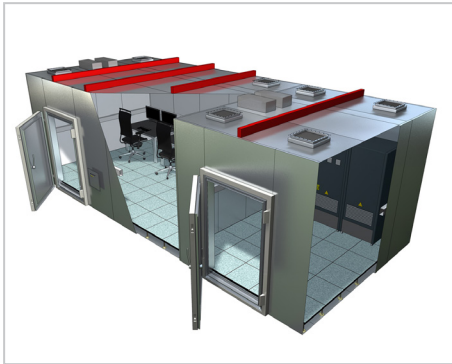
Modular and Prefabricated PAN Type Shielding

Frankonia shielded rooms and special security chambers are designed based on a modular construction system. Prefabricated high quality shielding panels guarantee a maximum of flexibility regarding possible dimensions.

All PAN type modules allow an easy handling and entry via standard building doors. The standard modules are bolted from inside every 75 mm with high conductivity mesh gasket inserted for sealing the joints of the panels. This facilitates an installation close to the walls of the parent building. The short screwing distance and the precise tightening of the screws with predefined torque guarantee long life shielding attenuation characteristics.

Features

- › PAN Type shielding modules made of 2,0 mm thick galvanized steel
- › Modular and prefabricated standard
- › Self-supporting stability or with static steel structure for any seismic condition
- › Mounted from the inside
- › Raised floor systems, or welded floor systems
- › Long life shielding attenuation characteristics
- › No glue, no welding
- › Dismountable without any damage, easy modifications and maintenance
- › A complete transfer or future modification is possibility
- › Turnkey solution



Shielding Standards

- › Frequency range acc. to EN 50147-1 or IEEE 299 (option) from 10 kHz up to 18 GHz or 40 GHz
- › Equal performance for any kind of feed-through components, honeycombs, doors and gates, filters, etc.
- › Any size of shielding is possible
- › Acoustic panels with absorption per ISO 354 $\alpha_w=0,65$ (MH)

Guaranteed Performance (min.)

Frequency	Attenuation	Field
10 kHz	≥ 90 dB	Magnetic Field
100 kHz	≥ 100 dB	Magnetic Field
1 MHz	≥ 110 dB	Magnetic Field
100 MHz	≥ 120 dB	Plane Wave
400 MHz	≥ 120 dB	Plane Wave
1 GHz	≥ 110 dB	Plane Wave
18 GHz	≥ 100 dB	Microwave
40 GHz	≥ 100 dB	Microwave

Shielding Components & Accessories

Shielding, Doors, Filters, Fluid feed-through and Accessories

Since 1987, Frankonia follows a prefabrication and modular standard at highest quality and efficiency. Nothing is welded, nothing is glued, everything remains modular to meet any future modification requirements.

Shielding and Structure

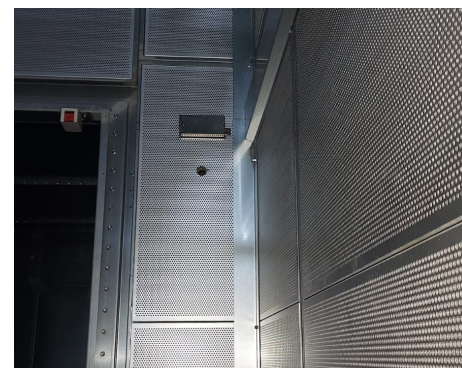
- › Modular and prefabricated PAN type shielding system
- › Highest shielding attenuation for every shielding accessories, honeycombs, doors, feed-through elements, electrics, and gates
- › Acoustic panels (F.DJ-T) absorption per ISO 354
- › Advanced facility isolated construction methods

Doors and Gates

- › Broad range of doors and gates in various sizes
- › Modular and prefabricated system
- › Single-leaf door (SLD)
- › Double-leaf door (DLD)
- › Sliding door (SSD) or gates (SG)

Ramps and Platforms

- › Automatic ramps and platforms (flush entrance)
- › Customized entrance solutions
- › Staff and material sluice systems



As a specialist in RF-shielding, Frankonia offers a broad range of complementary, standardized and customized products to maintain its position as a turnkey provider.

Every room is custom designed with its own full-integrative electrical system setup, simple interfaces to the facility, with additional and individual accessories.

Electrical Integration and Compliance

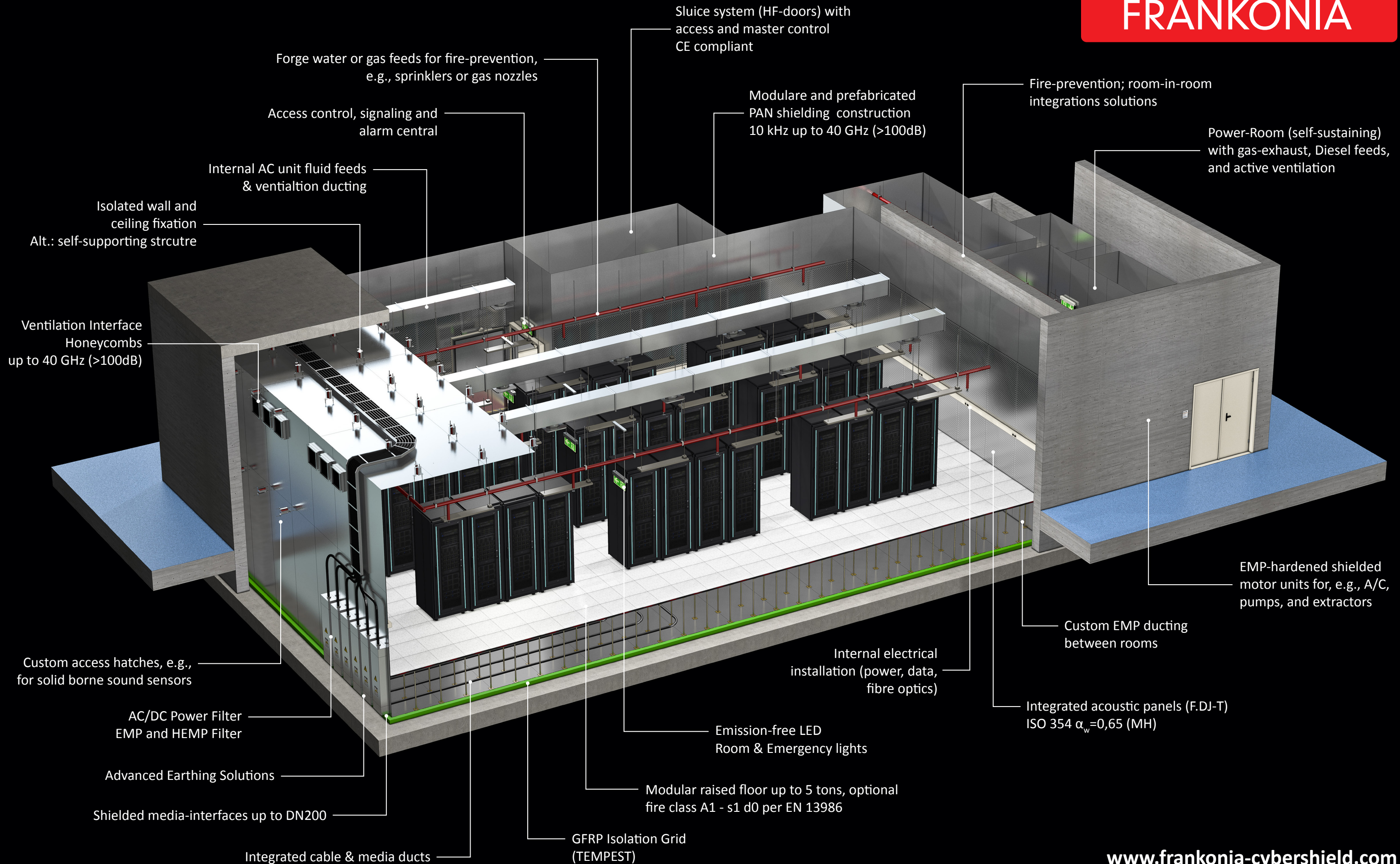
- › Electrical distribution unit (outside accessible), cabling, safety functions per local standards
- › LED lighting, explosion protection option, emergency lighting
- › AC and DC filters, signal and data filters, optic converters
- › Safety matrix & higher-level laboratory control unit (PLC system)
- › CE conformity as per Machinery Directive 2006/42/EG for product compliance as standard, or for the complete laboratory as option, paired with advanced safety measures

Ventilation, Fluid, feed-through, Smoke and Gas

- › Honeycombs, fluid feed-through's, cooling and exhaust systems
- › Air sampling network for gas and smoke detection systems
- › Smoke and gas detection analyzer with alarm central; ATEX compliance
- › Liquid detection system
- › Extinguishing solutions (Sprinklers, etc.)

Video & Audio Systems

- › HD camera systems, fixed or mobile version
- › Audio systems
- › Recording systems



Customized



End-to-end Frankonia Solutions

Comprehensive Products and Turnkey Portfolio

From first contact to verified protection

As a specialist in RF shielding, Frankonia offers a broad range of complementary, standardized and customized products to maintain its position as a turnkey provider. Every room is custom designed with its own full-integrative electrical system setup and simple interfaces to the facility.

- 01

Initial consultation
A specialist reviews your goals, site and sontraints. No documentation is required to start.
- 02

Risk & site assessment
Assets, threat scenarios, ambietn RF conditions and facility constraints are assessed; protection requirements are defined together.
- 03

Concept & 3D engineering
You receive a concept design, CAD achritectual integration and a transparent quotation.
- 04

Precision manufacturing
PAN mnodules, RF doors, power and data fitlers , medai feed throughs and honeycomb vents are produced in Frankonia's own facilities.
- 05

Installation & verification
Certified teams assemble the modules without welding; shielding effectiveness and leak detection are measures and documented at handover.
- 06

Operation & support
Preventive maintenance, periodic re-testing and security re-certification keep the boundary effective over time.

Complete scope, one supplier

Electrical Integration

- › Distribution unit, cabling, safety functions per local standards
- › LED lighting, explosion protection option, emergency lighting
- › AC and DC filters, signal and data filters, optic converters
- › Safety matrix and higher-level control unit (PLC)
- › CE conformity per Machinery Directive 2006/42/EG

Ventilation, Media Feeds, Smoke and Gas

- › Honeycombs, media feed-through's, cooling and exhaust systems
- › Air sampling network for gas and smoke detection
- › Detection analyzer with alarm central; ATEX compliance
- › Liquid detection system
- › Extinguishing solutions (sprinklers, etc.)



Frankonia, and where partners get involved

A transparent split of responsibilities across the protected room. Nothing inside the boundary is left to assumption.

Category	Importance	Scope & Expertise
RF shielding performance	CORE	In-house manufacturing with guaranteed performance
Doors & access solutions	CORE	Precision RF hinged and sliding doors
Electric filters	CORE	Power, data and signal RF filters
Ventilation & Fluid	CORE	Honeycombs, fluid media feed-throughs, ventilation systems
EMC testing & certification	CORE	On-site validation and leak detection
Maintenance & services	SERVICE	Annual audits and lifecycle support
Building security systems	PARTNER	Interface integration with facility security
Intrusion alarm systems	PARTNER	Interface integration with room alarm systems
Organizational procedures	CUSTOMER	Best-practice advisory provided by Frankonia

Advantages & Benefits

What makes our uniqueness and the perfect match

Frankonia is a top-notch solution provider for EMC Solutions from Germany. We offer a well-developed and an all-encompassing range of distinctive products to customize every project with perfection and time.

Product Advantages when selecting Frankonia

- › Modular and self-supporting system guarantees the highest shielding performance
- › Prefabricated standard paired with in-house production assures best quality
- › Extremely broad range of standard products and custom-specific products
- › In-house developed products are up to date considering latest technologies
- › Identical shielding performance throughout the whole system
- › Nothing is glued or welded; everything is mounted using bolting and screwing techniques
- › No timber works at all
- › No harmful materials at all (e.g., no carbon, no polyethylene, no glue)
- › Local adjustments to electricity, statics, technical parameters or building conditions are self-evident
- › An all-encompassing product line secures a single-point solution provider
- › All products are part of a well-developed system perfectly matched and comply with industrial standards

Customer Benefits when selecting Frankonia

- › Proven technologies and solutions since more than 35 years
- › Just recyclable materials in use; everything produced in our modern and eco-friendly manufacturing
- › Overall, little building conditions and requirements to consider
- › Optimized design approaches for an easy and fast installation
- › Made-to-order arrangement of chamber with accessories that fits into your building
- › Simple interfaces to the building (electricity, ventilation, exhaust, data, gas, ...)
- › Modular setup to re-located or modify
- › Experienced project management cares from the tiny details to the big picture completing on point
- › Professional installation service and after-sales service by Frankonia

Why customers selecting Frankonia

- › Require perfection and reliability to all individual needs and for any project sizes
- › Consider future-proven and long-lasting products from a single source
- › Need building design support with cost-saving efforts
- › Maximizing productivity and efficiency with optimal use of resources
- › Desire to get an outstanding lab at highest satisfaction

No compromises; just satisfaction throughout every custom-built setup. Frankonia provides authentic, proven and reliable solutions for the top-notch customer segment.

Frankonia Core Values

- › Ingenuity
We never stop pushing forward
We look for advantage in everything
We handle challenges with creativity
We strive for simplicity
- › Energy
We bring energy to everything we do
We stay hungry and believe in what we do
We are open to challenges and embrace opportunities
We share the passion for what we do
- › Community
We act fair and respectful
We roll up our sleeves to get the job done
We shape the future together
We win as a team
- › Fairness
We are open minded and curious
We admit failures, learn and improve
We communicate with clarity and honesty
We take responsibility for our actions

Sales & Information

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Frankonia EMC Test-Systems GmbH

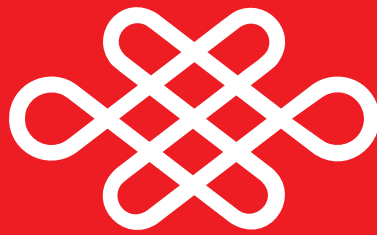
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FRANKONIA

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